

Fairness Is Advantage-Neutral: Guided Quantum Gibbs Dynamics for Fair Allocation of Indivisible Resources

Ritesh Roshan

School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Faridabad, Haryana, India

Abstract

Fair allocation of scarce indivisible resources is a natural target for quantum optimisation, but fairness criteria such as max-min welfare and the Gini coefficient cannot be written as quadratic binary objectives without auxiliary variables. We show that this obstruction disappears once allocations are randomised. By convex duality every concave fairness criterion reduces to a weighted-welfare problem whose Hamiltonian has the degree of the valuations, at zero auxiliary cost, and ex-ante fair allocation is polynomially equivalent to weighted-welfare allocation on any computational model. Fairness is therefore advantage-neutral: a quantum speedup for fair allocation exists exactly when one exists for the underlying combinatorial problem. Repeated fair allocation becomes a multiplicative-weights game whose rounds are Gibbs sampling problems with slowly varying targets. We prove a regret guarantee and a warm-start theorem under which each round costs a number of chain steps set by the spectral gap alone. For fair scheduling under interference we construct a quantum-enhanced Metropolis sampler whose proposals come from a palindromic Trotterisation of a cost Hamiltonian plus the blockade-preserving PXP driver. We prove that the circuit never leaves the feasible set, that the proposal is exactly symmetric at every Trotter depth, and that the chain converges to the fair Gibbs lottery. We give the circuit at gate level with closed-form resource counts and verify it against exact simulation to 10^{-15} . On three graph families with up to twenty links, quantum proposals enlarge the spectral gap over local moves by factors of 3.4 to over 1000, with geometric means of 20 to 49 per family. On Erdős–Rényi graphs the fitted gap-scaling exponent improves on the best classical baseline by a factor of 1.9 ± 0.5 to 2.3 ± 0.8 , but not on unit-disk or random 3-regular graphs. In repeated max-min allocation the quantum-mixed chain reaches a fairness value of 1.327 ± 0.007 against 1.179 ± 0.139 classically, with ex-ante optimum 1.460. These results establish constant-factor and instance-dependent gains, not a general quantum advantage.

Keywords: fair division; quantum optimisation; Markov chain Monte Carlo; convex duality; independent sets; Rydberg arrays

1 Introduction

Allocating food, vaccines, ventilators, spectrum or transmission slots under scarcity is an equity problem as much as an efficiency problem. Welfare economics offers many ways of stating what a fair allocation is, from Rawlsian max-min and leximin through the Gini and Atkinson families to Nash welfare and envy-based criteria [42, 4, 32, 23]. Quantum optimisation hardware accepts a much narrower input: a quadratic objective over binary variables, or its Ising equivalent. Max-min welfare has multilinear degree up to the number of agents, and absolute differences, and hence the Gini coefficient, have degree at least three.

Neither can be written as a quadratic binary objective without auxiliary variables and penalties [11]. Published quantum treatments of equity therefore either substitute a quadratic surrogate for the intended criterion [54] or move the fairness objective out of the Hamiltonian altogether [7].

This paper starts from a different observation. When goods are indivisible, the object of fair allocation is a lottery over allocations rather than a single allocation [20, 24], and randomised rationing is already used for scarce medical resources [56, 41]. Over lotteries the attainable utility set is convex. Convex duality then separates the normative content of a fairness criterion from the combinatorial content of the feasible set, and this separation settles both the representation question and the question of where a quantum device can help.

Contributions

1. A normative–combinatorial separation theorem (Theorem 2). Every concave fairness criterion, together with any requirement linear in the lottery, reduces to weighted-welfare best responses at zero auxiliary cost (Corollary 3).
2. Advantage neutrality (Theorem 4): ex-ante fair allocation and weighted-welfare allocation are polynomially equivalent on every computational model.
3. A dynamic formulation of repeated fair allocation with a regret guarantee (Theorem 9) and a warm-start theorem (Theorem 11) that bounds the per-round sampling cost by the spectral gap, with no dependence on the size of the state space.
4. A quantum proposal for fair scheduling under interference that preserves feasibility exactly (Lemma 12) and is exactly symmetric at every Trotter depth (Lemma 14), yielding a chain with the fair Gibbs lottery as its unique stationary distribution (Theorem 15). The circuit is given at gate level with closed-form counts.
5. An open, verified implementation and an experimental study on three graph families with five classical and quantum kernels, including exact warm-start tracking costs along multiplicative-weights trajectories.

The results are deliberately scoped. The theory shows that fairness cannot be the source of a quantum advantage. The experiments show where constraint-native quantum proposals help a fair sampler and where they do not. On present evidence, no general quantum advantage is claimed.

2 Related work

Fairness in allocation. The axiomatics of inequality measurement go back to the transfer principle [18] and the equally-distributed-equivalent framework [4]. The generalized Lorenz order is the consensus of Schur-concave welfare functions [45, 38]. The informational basis of social choice determines which aggregators are admissible [28, 19, 32], and the price of fairness quantifies the efficiency cost of equity [10]. For indivisible goods, envy-freeness up to one good always exists [36, 14], maximin shares need not [33], and envy-freeness up to any good fails for monotone and submodular valuations [1]. Randomisation recovers exact ex-ante fairness together with approximate ex-post fairness [24]. Ex-ante envy-freeness combined with Pareto optimality is a PPAD problem [15]. Minimising the sum of variances over lotteries, the

quadratic-native alternative, is NP-hard and does not guarantee ex-post EF1 for non-identical valuations [5].

Quantum optimisation and encodings. Quadratic binary encodings are catalogued in [37]. Constraint-preserving mixers remove equality penalties [27, 55]. Low-depth variational optimisation faces locality and symmetry obstructions [22, 13], and quadratic speedups do not survive error-correction overheads [6]. Superpolynomial speedups require structure: algebraic structure in decoded quantum interferometry [31, 3], planted structure in guided Hamiltonian problems [44], and engineered adiabatic paths without a sign problem [26, 29]. Maximum independent set is native to Rydberg arrays through the blockade mechanism [9, 21], although classical solvers remain competitive on unit-disk instances [2].

Quantum sampling. Quantum walks give quadratic improvements in spectral gap [51, 47]. Quantum-enhanced Markov chain Monte Carlo uses measured quantum evolution as a proposal, with classical acceptance ensuring the correct stationary distribution [34]. On worst-case unstructured problems, unital quantum proposals give no speedup [40]. Dynamic Gibbs sampling underlies quantum speedups for zero-sum games [12]. To our knowledge, fairness criteria have not previously been placed in this sampling framework.

3 Problem formulation

3.1 Setting

Let $[n]$ be agents and $F \subseteq \{0, 1\}^M$ a finite set of feasible allocations. A statistic map $g : F \rightarrow \mathbb{R}^D$ contains the utilities $u_i(x) \geq 0$ among its coordinates, with $\rho = \max_{i,x} u_i(x)$. The attainable image is $U = g(F)$, a lottery is $p \in \Delta(F)$, and its mean is $\mathbb{E}_p[g] = \sum_x p(x)g(x)$. The set of lottery means is $P = \text{conv } U$, since every convex combination of points of U is the mean of the corresponding lottery. We write $h_U(\theta) = \max_{x \in F} \theta \cdot g(x)$ for the support function and $\Phi_*(\theta) = \inf_y \{\theta \cdot y - \Phi(y)\}$ for the concave conjugate.

3.2 Fairness criteria

Table 1 lists the principal criteria, their definitions, and the admissible priority weights obtained from their conjugates. Derivations are given in Appendix A. Throughout, $y_{(1)} \leq \dots \leq y_{(n)}$ denote order statistics and μ the mean.

Definition 1 (FAIR). Given (F, g) , a proper closed concave $\Psi : \mathbb{R}^D \rightarrow [-\infty, \infty)$ that encodes the welfare criterion and any convex requirements, and $\varepsilon > 0$, find a lottery $p \in \Delta(F)$ with $\Psi(\mathbb{E}_p[g]) \geq \sup_q \Psi(\mathbb{E}_q[g]) - \varepsilon$.

4 Structural results

Theorem 2 (Normative-combinatorial separation). For every proper closed concave Ψ ,

$$\max_{p \in \Delta(F)} \Psi(\mathbb{E}_p[g]) = \inf_{\theta} [h_U(\theta) - \Psi_*(\theta)],$$

and $g(x^*(\theta)) \in \partial h_U(\theta)$ for every $x^*(\theta) \in \arg \max_{x \in F} \theta \cdot g(x)$.

Proof. The left side equals $\max_{y \in P} \Psi(y)$. By the Fenchel–Moreau theorem [43], $\Psi(y) = \inf_{\theta} \{\theta \cdot y - \Psi_*(\theta)\}$. The set P is compact and convex, and the function $\theta \cdot y - \Psi_*(\theta)$ is affine in y and

Criterion	Definition	Dual weights θ
Utilitarian	$\sum_i y_i$	$\theta = \mathbf{1}$
Max-min	$\min_i y_i$	$\theta \in \Delta_n$
Leximin	lexicographic on $(y_{(1)}, \dots, y_{(n)})$	nested max-min stages
Lorenz component	$L_k(y) = \sum_{j \leq k} y_{(j)}$	$0 \leq \theta \leq 1, \sum_i \theta_i = k$
Ordered weighted average	$\sum_k \omega_k y_{(k)}, \omega$ decreasing	permutohedron of ω
Gini–Sen welfare	$\mu(1 - G) = \sum_k \frac{2(n-k)+1}{n^2} y_{(k)}$	permutohedron of Gini weights
Nash	$\sum_i \log y_i$	$\theta_i = 1/y_i$ at optimum
α -fair, Atkinson	$\sum_i y_i^{1-\alpha} / (1 - \alpha)$	$\theta_i = y_i^{-\alpha}$ at optimum
Kalai–Smorodinsky	$\min_i (y_i - d_i) / (a_i - d_i)$	simplex, rescaled
Need floor, proportionality	$y_i \geq d_i, y_i \geq \frac{1}{n} \mathbb{E}W$	multipliers $\mu_i \geq 0$
Ex-ante envy-freeness	$\mathbb{E}u_i(x_i) \geq \mathbb{E}u_i(x_j)$	multipliers $\mu_{ij} \geq 0$
Mean minus variance	$\mu - \lambda \text{Var}(y)$	weights may be negative

Table 1: Fairness criteria and their dual priority weights.

convex and lower semicontinuous in θ , so Sion’s theorem [46] exchanges $\max_y$ and $\inf_\theta$. A linear function attains its maximum over a polytope at a vertex, so $\max_{y \in P} \theta \cdot y = h_U(\theta)$. The support function is a pointwise maximum of affine functions, and the gradient of an active piece is a subgradient.

Corollary 3 (Zero representation cost). *The inner problem $\max_{x \in F} \theta \cdot g(x)$ has the multilinear degree of g for every concave Ψ . No fairness criterion in Table 1 requires auxiliary variables beyond those that define F .*

Theorem 4 (Advantage neutrality). *On every computational model, ex-ante FAIR and weighted-welfare optimisation over the same feasible family are polynomially equivalent.*

Proof. A first-order method on the dual of Theorem 2 reaches accuracy ε in $\text{poly}(D, \rho, 1/\varepsilon)$ iterations, each needing one approximate weighted best response [25]. Conversely, $\Psi(y) = \theta \cdot y$ is an instance of FAIR; its optimum over P is attained at a vertex, so a sparse optimal lottery contains an optimal allocation in its support.

Proposition 5 (Where fairness itself is hard). *Deterministic max-min allocation is NP-hard for two agents with identical additive valuations, by reduction from PARTITION, whereas the ex-ante problem is solved by a fair coin. For additive valuations over m items, every $y \in P$ is the utility vector of a fractional allocation with at most n split items, so a deterministic allocation achieves $u_i \geq y_i - nV_{\max}$ for all i .*

Proof. The second claim follows from the Shapley–Folkman theorem [48] applied to $U = \sum_j S_j$ with $S_j = \{0\} \cup \{v_{ij}e_i\}_i$: all but n summands of any point of $P = \sum_j \text{conv } S_j$ are extreme, and rounding each split item costs each agent at most $V_{\max}$ per split item.

Proposition 6 (Representability duality). *Max-min and Gini-type criteria have high per-allocation degree but are concave over lotteries. Sum-of-variances has per-allocation degree $2d$, but $\sum_i \text{Var}_p(u_i) = \mathbb{E}_p[\sum_i u_i^2] - \sum_i (\mathbb{E}_p u_i)^2$ is concave in p , so its minimisation over lotteries is a concave minimisation.*

Proof. The degree statements follow from the multilinear expansion [11]. With indicator utilities $u_i = x_i$, $\min_i u_i = \prod_i x_i$ has degree n . With $u_1 = x_1 + x_2 + x_3$ and $u_2 = 1, |u_1 - u_2|$

has cubic coefficient -2 . Squares of linear forms have degree two. The concavity claim holds because the second term is minus a convex quadratic of a linear map of p .

Lemma 7 (Gibbs variational principle). *For $E : F \rightarrow \mathbb{R}$ and $\pi_\beta \propto e^{\beta E}$, $\mathbb{E}_{\pi_\beta}[E] \geq \max_x E(x) - \log |F| / \beta$.*

Proof. For any q , $\beta \mathbb{E}_q[E] + H(q) - \log Z = -\text{KL}(q \| \pi_\beta) \leq 0$, with equality at $q = \pi_\beta$. Comparing with a point mass at a maximiser gives $\beta \mathbb{E}_{\pi_\beta}[E] + H(\pi_\beta) \geq \beta \max E$, and $H \leq \log |F|$.

Theorem 8 (The fairest lottery is a generalized Gibbs state). *For $y \in \text{relint } P$, the maximum-entropy lottery with mean y is $p^*(x) = e^{\theta^* \cdot g(x)} / Z(\theta^*)$, where θ^* minimises the convex function $\log Z(\theta) - \theta \cdot y$.*

Proof. Stationarity of the Lagrangian $H(p) + \theta \cdot (\mathbb{E}_p[g] - y) + \nu(\sum_x p(x) - 1)$ gives the exponential form, and strict concavity of entropy gives uniqueness [30].

5 Dynamic fair allocation

In rounds $t = 1, \dots, T$, priority weights $\theta_t \in \Delta_n$ evolve by

$$\theta_{t+1,i} = \frac{\theta_{t,i} e^{-\eta u_i(x_t)}}{\sum_j \theta_{t,j} e^{-\eta u_j(x_t)}}, \quad \theta_1 = \frac{1}{n} \mathbf{1}, \quad (1)$$

and each allocation is drawn from the Gibbs target $\pi_t(x) \propto e^{\beta n \theta_t \cdot u(x)}$ on F . The factor n normalises the weights to unit mean.

Theorem 9 (Dynamic fairness guarantee). *Let $v^* = \max_p \min_i \mathbb{E}_p u_i$ and $\eta = \rho^{-1} \sqrt{8 \ln n / T}$. If each x_t is drawn within total variation ε of π_t , then*

$$\mathbb{E} \left[\min_i \frac{1}{T} \sum_{t=1}^T u_i(x_t) \right] \geq v^* - \frac{\log |F|}{\beta n} - \rho \varepsilon - \rho \sqrt{\frac{\ln n}{2T}}.$$

Proof. By Theorem 2 with $\Psi = \min_i, v^* = \min_{\theta \in \Delta_n} h_U(\theta) \leq h_U(\theta_t)$. Lemma 7 with $E = n \theta_t \cdot u$ and inverse temperature β gives $\mathbb{E}_{\pi_t}[\theta_t \cdot u] \geq h_U(\theta_t) - \log |F| / (\beta n)$, and a sampling error ε costs at most $\rho \varepsilon$. For Hedge with losses in $[0, \rho]$, Hoeffding's lemma gives $\ln(W_{t+1}/W_t) \leq -\eta \theta_t \cdot u(x_t) + \eta^2 \rho^2 / 8$. Summing, $\sum_t \theta_t \cdot u(x_t) - \min_i \sum_t u_i(x_t) \leq \ln n / \eta + T \eta \rho^2 / 8 = \rho \sqrt{T \ln n / 2}$ pathwise [16]. Taking expectations and dividing by T gives the claim.

Lemma 10 (Target drift). *Under (1), $\|\theta_{t+1} - \theta_t\|_1 \leq e^{\eta \rho} - 1$. With $\delta_t = \max_x |(\theta_{t+1} - \theta_t) \cdot u(x)|$, $e^{-2\beta n \delta_t} \leq \pi_{t+1}(x) / \pi_t(x) \leq e^{2\beta n \delta_t}$.*

Proof. Each ratio $\theta_{t+1,i} / \theta_{t,i}$ lies in $[e^{-\eta \rho}, e^{\eta \rho}]$. The target ratio is $e^{\beta n (\theta_{t+1} - \theta_t) \cdot u(x)} Z_t / Z_{t+1}$, where both factors lie in $[e^{-\beta n \delta_t}, e^{\beta n \delta_t}]$.

Theorem 11 (Warm-start tracking). *Let P_{t+1} be reversible with stationary distribution π_{t+1} and absolute spectral gap at least γ . If the chain starts from μ with $\chi^2(\mu \| \pi_t) \leq \varepsilon^2$, then after*

$$k \geq \frac{1}{2\gamma} \ln \frac{e^{2\beta n \delta_t} (1 + \varepsilon^2) - 1}{\varepsilon^2}$$

steps, $\chi^2(\mu P_{t+1}^k \| \pi_{t+1}) \leq \varepsilon^2$ and the total variation distance is at most $\varepsilon / 2$. The bound does not depend on $|F|$ or on $\min_x \pi_t(x)$.

Proof. By Lemma 10, $1 + \chi^2(\mu \| \pi_{t+1}) = \sum_x \mu(x)^2 / \pi_{t+1}(x) \leq e^{2\beta n \delta_t} (1 + \chi^2(\mu \| \pi_t))$. For a reversible chain, $\mu / \pi - 1$ is orthogonal to constants in $L^2(\pi)$ and contracts by $1 - \gamma$ per step [35], so χ^2 contracts by $(1 - \gamma)^{2k} \leq e^{-2\gamma k}$. Finally, $\text{TV} \leq \frac{1}{2} \sqrt{\chi^2}$.

The cost of fair allocation is therefore governed by the spectral gap of the sampler, which is the quantity a quantum proposal can change.

6 The quantum proposal

6.1 Instance class

Fair scheduling under interference takes a conflict graph $G = (V, E)$ on N links, with rates $r_v > 0$ and owners $a : V \rightarrow [n]$. Feasible schedules are independent sets, $F = \mathcal{I}(G)$, and $u_i(x) = \sum_{v:a(v)=i} r_v x_v$. With vertex weights $w_v = n \theta_{a(v)} r_v$, the target is $\pi(x) \propto e^{\beta \sum_v w_v x_v}$, and the best-response problem is maximum-weight independent set, the per-slot problem of max-weight scheduling [52].

6.2 Hamiltonians and invariance

Let $n_v = |1\rangle\langle 1|_v$, $P_v^0 = |0\rangle\langle 0|_v$ and $\Pi_v = \prod_{u \in N(v)} P_u^0$, and define

$$H_\theta = - \sum_v w_v n_v, \quad H_{\text{PXP}} = \sum_v X_v \Pi_v, \quad H = \gamma_c H_\theta + h H_{\text{PXP}}.$$

Lemma 12 (Subspace preservation). *Every factor e^{-isH_θ} and $e^{-i\alpha X_v \Pi_v}$, and every product of such factors, preserves $\mathcal{H}_F = \text{span}\{|x\rangle : x \in F\}$ exactly.*

Proof. For $x \in F$, $\Pi_v |x\rangle$ is nonzero only if every neighbour of v is empty, in which case toggling x_v yields an independent set. The cost Hamiltonian is diagonal.

Lemma 13. *If u and v are non-adjacent, then $[X_u \Pi_u, X_v \Pi_v] = 0$.*

Proof. Each Pauli acts on a site where the other term's projector is the identity, and the projectors are diagonal.

Given a proper colouring $C_1, \dots, C_\chi$ and $\tau = t_{\text{ev}}/K$, one palindromic step is

$$S(\tau) = \mathcal{C}(\frac{\tau}{2}) \mathcal{D}_1(\frac{\tau}{2}) \cdots \mathcal{D}_{\chi-1}(\frac{\tau}{2}) \mathcal{D}_\chi(\tau) \mathcal{D}_{\chi-1}(\frac{\tau}{2}) \cdots \mathcal{D}_1(\frac{\tau}{2}) \mathcal{C}(\frac{\tau}{2}), \quad (2)$$

with $\mathcal{C}(s) = e^{-is\gamma_c H_\theta}$ and $\mathcal{D}_c(s) = \prod_{v \in C_c} e^{-ish X_v \Pi_v}$. Each $\mathcal{D}_c$ is exact by Lemma 13, and the proposal unitary is $\mathcal{U} = S(\tau)^K$ [49, 50].

Lemma 14 (Exact symmetry). $\mathcal{U}^\top = \mathcal{U}$ in the computational basis. Hence $Q(x | y) = |\langle x | \mathcal{U} | y \rangle|^2$ is symmetric for every K and τ .

Proof. Every generator is real symmetric, so every factor equals its own transpose. The transpose of a palindromic product is the same product, and powers of a symmetric matrix are symmetric.

Theorem 15 (Correctness). *Let $\tilde{Q} = (1 - q)K_{\text{loc}} + qQ$ with K_{loc} the single-vertex flip proposal and $q < 1$. The lazy Metropolis kernel $P(y, x) = \frac{1}{2} \tilde{Q}(x | y) \min\{1, \pi(x)/\pi(y)\}$ for $x \neq y$ is reversible with respect to π , irreducible and aperiodic on F , and has nonnegative spectrum.*

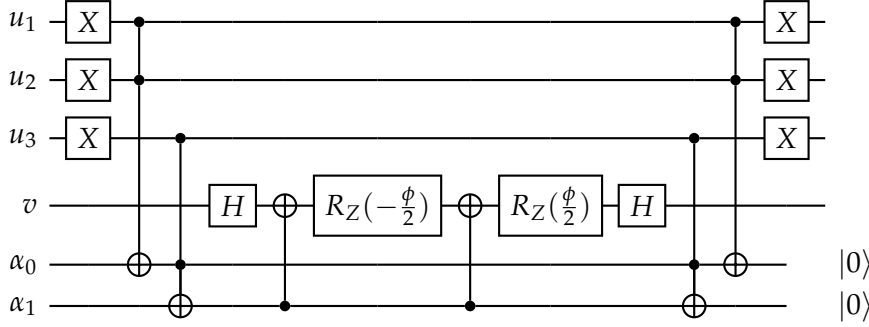
Proof. Both proposals are symmetric (Lemma 14), so $\pi(y)P(y, x) = \frac{1}{2}\tilde{Q}(x | y) \min\{\pi(x), \pi(y)\}$ is symmetric. Every independent set connects to $\emptyset$ by single flips. Laziness gives aperiodicity and eigenvalues in $[0, 1]$.

6.3 Circuit

The following identities give every gate:

$$\begin{aligned} e^{-is\gamma_c H_\theta} &= \prod_v P(s\gamma_c w_v), & P(\phi) &= \text{diag}(1, e^{i\phi}), \\ e^{-i\alpha X_v \Pi_v} &= R_X(2\alpha)_v \otimes \Pi_v + I_v \otimes (I - \Pi_v), \\ CR_X(\phi) &= (I \otimes H)(I \otimes R_Z(\frac{\phi}{2}))\text{CNOT}(I \otimes R_Z(-\frac{\phi}{2}))\text{CNOT}(I \otimes H). \end{aligned}$$

The first follows from $n_v = |1\rangle\langle 1|$, the second from $\Pi_v^2 = \Pi_v$ and $X^2 = I$, and the third from $HR_ZH = R_X$ and $XR_ZX = R_Z^\dagger$. Open controls are implemented by conjugating each neighbour with X , since $XnX = P^0$. The conjunction of $d_v \geq 2$ controls is computed into $d_v - 1$ ancillas by a chain of Toffoli gates and uncomputed afterwards [8, 39]. For three neighbours:



Let $m_v = 1$ if $v \in C_\chi$ and $m_v = 2$ otherwise. Per Trotter step the circuit uses exactly

$$\begin{aligned} n_P &= 2N, & n_X &= \sum_v 2m_v d_v, & n_{CCX} &= \sum_v 2m_v (d_v - 1)^+ \leq 8|E|, \\ n_{\text{CNOT}} &= n_H = n_{R_Z} = \sum_{v: d_v \geq 1} 2m_v, \end{aligned}$$

with one unconditional R_X per isolated vertex application and $N + \max(\Delta - 1, 0)$ qubits. For the benchmark unit-disk instance ($N = 20$, $|E| = 53$, $\Delta = 10$, $\chi = 6$; Appendix B) with $t_{\text{ev}} = 5$, $K = 10$, $\gamma_c = 1$ and $h = 0.4$, one step has 40 phase, 400 X , 324 Toffoli, 76 CNOT, 76 Hadamard and 76 R_Z gates, on 29 qubits with depth 445. After decomposing each Toffoli into six CNOT, two H , seven T or $T^\dagger$ and one S gate, a step has 2020 CNOT. On a Rydberg array in the blockade regime, H is the native Hamiltonian with local detunings $\Delta_v = \gamma_c w_v$ and Rabi frequency $\Omega = 2h$ [9, 53], and no gate compilation is needed.

6.4 Algorithm

Input: G, r, a, T, β, η , chain lengths k_0 and k , mixing probability q , and proposal parameters $\mathcal{P}$.
Output: a schedule $(x_t)_{t \leq T}$, its fairness value $\min_i \frac{1}{T} \sum_t u_i(x_t)$, and the dual bound $\min_t h_U(\theta_t)$.

- 1: colour G ; $\theta_1 \leftarrow \frac{1}{n}\mathbf{1}$; $y \leftarrow \emptyset$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: $w_v \leftarrow n\theta_{t,a(v)}r_v$
- 4: **for** $s = 1, \dots, k_t$ **do** $\triangleright k_1 = k_0$, else $k_t = k$

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5:         with probability  $\frac{1}{2}$  skip
6:         with probability  $q$ : prepare  $|y\rangle$ , apply  $S(\tau)^K$  with parameters drawn from  $\mathcal{P}$ , mea-
           sure  $x$ 
7:         otherwise: flip a uniform vertex of  $y$  if the result is independent, giving  $x$ 
8:         accept  $x$  with probability  $\min\{1, e^{\beta \sum_v w_v(x_v - y_v)}\}$ 
9:     end for
10:     $x_t \leftarrow y$ ; update  $\theta_{t+1}$  by (1)
11: end for

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The runtime is $O((k_0 + Tk)(N + qc_Q))$, where c_Q is the gate count of one proposal. The chain state persists across rounds, which is the warm start of Theorem 11.

7 Experiments

7.1 Setup

Three conflict-graph families were used:

- unit-disk graphs, with N points uniform in a square of area $N/2.5$, joined at distance at most one, for $N = 8, \dots, 20$;
- Erdős–Rényi graphs $G(N, 0.3)$, whose mean degree grows with N , for $N = 8, \dots, 18$;
- uniform random 3-regular graphs, for $N = 8, \dots, 18$.

Each size used five seeds. Rates are drawn from $U[0.5, 1.5]$ and three agents own the links in balanced random order; all instances are fully specified by their seeds. The feasible sets ranged from 17 to 2635 independent sets.

Five symmetric proposals were compared, each inside the same lazy Metropolis kernel:

- *local*: single-vertex flips;
- *local+swap*: single flips mixed equally with Kawasaki-type moves of a particle across an edge;
- *uniform*: uniform proposals over F , an idealised baseline, since exact uniform sampling of independent sets is intractable in general;
- *quantum*: an average over three parameter draws with $h \in [0.25, 0.6]$, $t_{\text{ev}} \in [2, 10]$, $\tau \leq 0.5$ and $\gamma_c = (1 - h)\sqrt{N}/\|w\|$, following [34];
- *mixed*: local and quantum with equal probability.

Proposals were computed by exact evolution restricted to $\mathcal{H}_F$, and spectral gaps by diagonalising the symmetrised kernel. Fixed-weight gaps use $w_v = r_v$ with $\beta \in \{1, 3\}$.

7.2 Verification

On five instances covering all three families, the gate-level circuit, simulated as a full state vector with ancillas, agreed with the subspace evolution to at most 3.1×10^{-15} , with leakage below 2.1×10^{-14} . The closed-form gate counts matched the generated circuits exactly. Proposal asymmetry was at most 1.7×10^{-16} , detailed-balance error at most 6.5×10^{-19} , and stationarity error at most 1.1×10^{-16} .

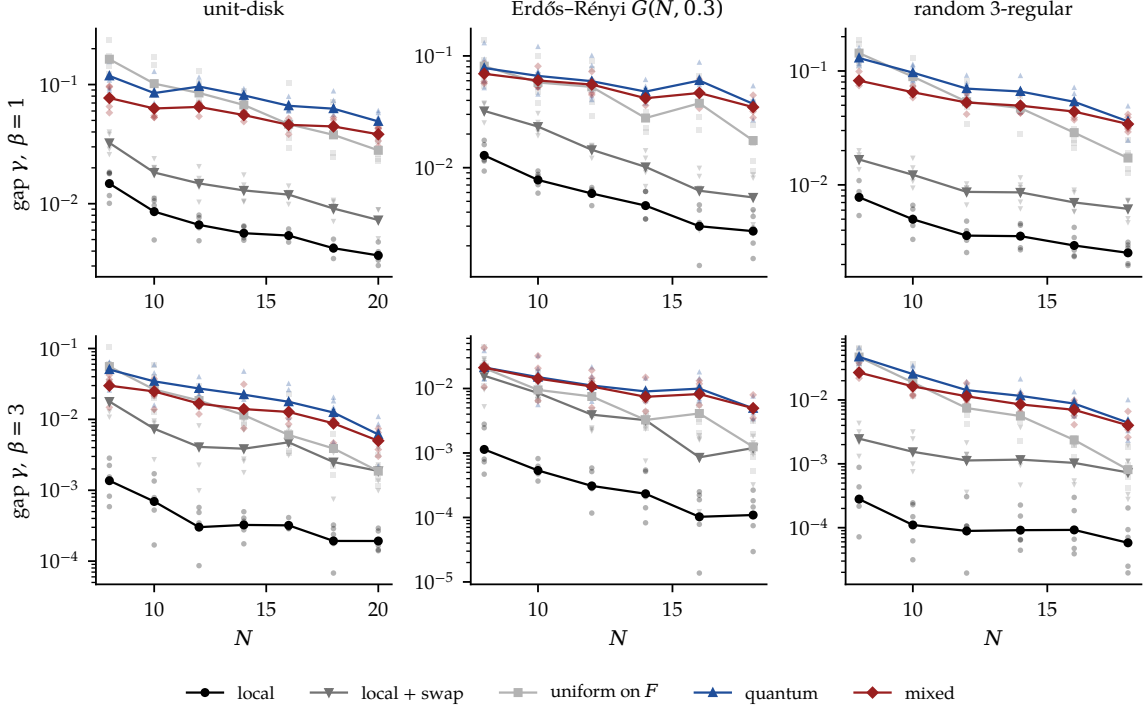


Figure 1: Spectral gaps of the five kernels against N for three graph families and two inverse temperatures. Points are individual instances and lines are geometric means over five seeds.

7.3 Spectral gaps

Figure 1 shows the gaps. Quantum proposals enlarge the gap over local moves in every instance, by factors between 3.4 and 1116; the geometric means are 25.6 on unit-disk, 20.3 on Erdős-Rényi and 48.6 on 3-regular graphs. The relevant question for advantage is scaling, so we fit $\log \gamma = -\kappa N + c$ and report the ratio $a = \kappa_C / \kappa_Q$ against the best classical kernel. This ratio is the exponent in the relation (classical steps) $\approx$ (quantum steps) ^{a} .

Family	β	κ_Q	best classical κ_C	$a = \kappa_C / \kappa_Q$
Unit-disk	1	0.065 ± 0.009	0.103 ± 0.010 (local)	1.60 ± 0.28
Unit-disk	3	0.157 ± 0.020	0.150 ± 0.026 (local)	0.95 ± 0.21
Erdős-Rényi	1	0.060 ± 0.017	0.137 ± 0.023 (uniform)	2.28 ± 0.76
Erdős-Rényi	3	0.125 ± 0.024	0.242 ± 0.042 (local)	1.94 ± 0.50
3-regular	1	0.118 ± 0.011	0.095 ± 0.011 (local+swap)	0.81 ± 0.12
3-regular	3	0.218 ± 0.020	0.104 ± 0.035 (local+swap)	0.48 ± 0.17

Table 2: Fitted gap-scaling rates with standard errors. Values of a above one favour the quantum kernel.

Table 2 shows that the scaling behaviour depends on the family. On Erdős-Rényi graphs the quantum kernel improves the scaling rate by a factor of about two. On unit-disk graphs it improves only at high temperature. On random 3-regular graphs the swap-augmented classical chain scales better than the quantum kernel.

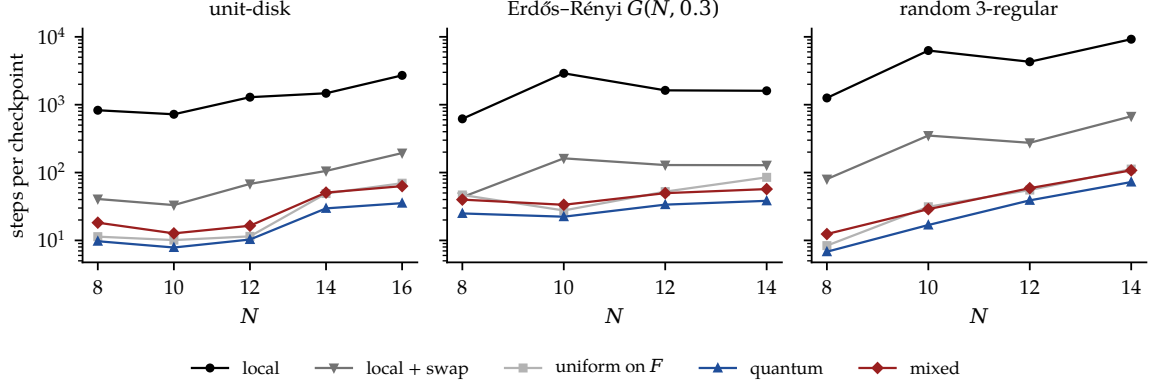


Figure 2: Exact warm-start tracking cost: the number of chain steps needed to move from π_t to within $\chi^2 \leq 0.01$ of π_{t+30} along multiplicative-weights trajectories ($T = 300, \beta = 3$). Values are geometric means over three seeds and nine checkpoints.

7.4 Tracking along fairness trajectories

Theorem 11 identifies the tracking cost as the operational quantity. For each instance we generated a multiplicative-weights trajectory with exact Gibbs sampling. For checkpoints thirty rounds apart, we computed exactly the least number of steps k with $\chi^2(\pi_t P_{t+30}^k \| \pi_{t+30}) \leq 0.01$. Figure 2 shows the results. The quantum kernel required the fewest steps in 82% of the 39 instances. In geometric mean it needed 21.4 steps against 32.3 for uniform proposals, 120.6 for local+swap and 1910 for local moves. Per instance, the best classical kernel required 1.44 times as many steps as the quantum kernel in geometric mean. The growth of the tracking cost with N is similar for the quantum and uniform kernels over the range tested.

7.5 Repeated max-min allocation

On the unit-disk instance with $N = 12$ and seed 1 ($|F| = 131$, three agents), the ex-ante optimum is $v^* = 1.460$ and the best single allocation achieves 1.227. With $\beta = 3, T = 1500, k = 8, k_0 = 200$ and five repetitions, Theorem 9 guarantees at least $v^* - 0.620$.

Chain	Fairness value	Gini of averages	Dual upper bound
classical ($q = 0$)	1.179 ± 0.139	0.0127	1.467
quantum-mixed ($q = 0.5$)	1.327 ± 0.007	0.0104	1.470

At equal step budget, the quantum-mixed chain exceeds the best deterministic allocation in every repetition and reduces the run-to-run standard deviation by a factor of about nineteen. The classical chain falls below the deterministic optimum in three of five runs.

8 Discussion

The theory fixes where a quantum device can contribute. Fairness adds no hardness ex ante, and no representation cost once lotteries are admitted (Theorems 2 and 4). Any speedup must therefore come from sampling the weighted feasible set, and the warm-start theorem makes the spectral gap the governing quantity.

The experiments show that constraint-native quantum proposals improve this quantity by

large constant factors in all families and reduce tracking cost in most instances. They improve the scaling rate on dense random graphs but not on sparse regular ones. For a reversible kernel, the spectral gap is at most twice the conductance of any set [35]. A quantum proposal therefore helps only by moving probability across the bottlenecks of π , and the family dependence observed here is consistent with the absence of a generic mechanism [40].

The Erdős–Rényi ratios of 1.9 ± 0.5 and 2.3 ± 0.8 are consistent with a quadratic-order improvement. They do not exceed the threshold that error-corrected advantage is estimated to require [6], and they are measured at small sizes.

Limitations.

- Instance sizes are limited to $N \leq 20$ by exact diagonalisation.
- Parallel tempering and cluster algorithms were not included, because their gaps are not computable exactly on the extended state space.
- Comparisons count chain steps rather than wall-clock cost, and a quantum step costs more than a classical flip.
- The uniform baseline is idealised.
- Hardware noise was not modelled. The proposal is exactly symmetric only for the ideal circuit, and noise-induced asymmetry would bias the stationary distribution unless mitigated.

Extending the study to $N \geq 30$ by matrix-free Lanczos methods, adding parallel tempering by autocorrelation measurements, testing non-stoquastic drivers [17], and costing proposals in physical time on Rydberg hardware would decide whether the Erdős–Rényi scaling persists.

9 Conclusion

Fair allocation of indivisible resources admits an exact reformulation in which fairness is a rule for updating priority weights and computation reduces to Gibbs sampling over the feasible set. This removes the representability obstruction that has shaped quantum treatments of equity. It also shows that fairness itself neither creates nor destroys quantum advantage. For fair scheduling under interference, a blockade-preserving quantum proposal gives an exactly correct sampler with gate-level resources stated in closed form. It yields large constant-factor improvements, lower tracking cost in most instances, and an improvement in gap scaling on dense random graphs, which is instance-dependent rather than general.

Declarations

Data availability. All data are generated by the released code from fixed seeds. The results files are included in the code archive.

Code availability. The complete implementation (`fairq`, version 1.0.0) with a one-command reproduction script is provided as supplementary material under the MIT licence.

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Author contributions. R.R. conceived the study, developed the theory, implemented the software, performed the experiments and wrote the manuscript.

Appendix A. Derivation of the dual weights

Max-min. For $\theta \in \Delta_n$, $\theta \cdot y \geq \min_i y_i$ with equality at $y = c\mathbf{1}$, so $\Phi_*(\theta) = 0$. Outside the simplex, taking $y = se_j$ with $s \rightarrow \infty$ (if $\theta_j < 0$) or $y = c\mathbf{1}$ with $c \rightarrow \pm\infty$ (if $\sum_i \theta_i \neq 1$) sends the infimum to $-\infty$.

Ordered weighted averages and Gini–Sen welfare. By the rearrangement inequality, $\sum_k \omega_k y_{(k)} = \min_\sigma \sum_i \omega_{\sigma(i)} y_i$ for decreasing ω , so the conjugate domain is the permutohedron. Moreover $\mu(1 - G) = n^{-2} \sum_{i,j} \min(y_i, y_j)$, and $y_{(k)}$ is the minimum of exactly $2(n - k) + 1$ ordered pairs.

Nash. $\inf_{y>0} (\theta y - \log y) = 1 + \log \theta$ at $y = 1/\theta$.

α -fairness. The first-order condition $\theta_i = y_i^{-\alpha}$ gives $\Phi_*(\theta) = \frac{\alpha}{\alpha-1} \sum_i \theta_i^{(\alpha-1)/\alpha}$.

Linear requirements. A requirement $a \cdot \mathbb{E}_p[g] \leq b$ adds a multiplier $\mu \geq 0$ times a to the priority weights by Lagrangian duality.

Pareto diagnostic. A closed concave Φ is nondecreasing if and only if its conjugate domain lies in $\mathbb{R}_{\geq 0}^n$. For mean minus variance, $\theta_i = \frac{1}{n} - \frac{2\lambda}{n}(y_i - \mu)$, which is negative for sufficiently well-off agents.

Appendix B. Benchmark instance

The benchmark is `make_instance(20, 3, 0, kind="udg")`. Its edges are

```
(0,5) (0,6) (0,7) (0,8) (0,12) (0,15) (0,17) (0,19) (1,10) (2,3) (2,4) (2,11)
(2,13) (2,14)
(2,18) (3,4) (3,8) (3,11) (3,12) (3,14) (3,15) (3,18) (4,11) (4,14) (4,18) (5,6)
(5,7) (6,7)
(6,19) (7,12) (7,15) (7,17) (7,19) (8,11) (8,12) (8,14) (8,15) (8,19) (9,12)
(9,16) (9,17)
(11,12) (11,14) (11,15) (11,18) (12,14) (12,15) (12,17) (12,19) (14,15) (14,18)
(15,17) (15,19)
```

and its colour classes are $C_1 = \{1,2,6,12,16\}$, $C_2 = \{4,5,9,10,13,15\}$, $C_3 = \{0,3\}$, $C_4 = \{7,11\}$, $C_5 = \{14,17,19\}$ and $C_6 = \{8,18\}$. The first driver gate of a step, for vertex 1 with $N(1) = \{10\}$, reads in OpenQASM 2.0

```
x q[10]; h q[1]; cx q[10],q[1]; rz(-0.1) q[1]; cx q[10],q[1]; rz(0.1) q[1]; h q
[1]; x q[10];
```

The full step is written by `circuit_spec.py` to `results/proposal_step.qasm`, and the rates and owners to `results/circuit_spec.json`.

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